POKHARA UNIVERSITY

Level: Bachelor | Semester: Fall | Year : 2024
Programme: BE | Full Marks : 100
Course: Problem Solving Techniques (New) | Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Determine how many zeros end the number $300! + 6415 \times 125^{10}$. 5

   b) What is the last digit of 7^{4563}? 5

   c) Suppose there are k students in the class.If K is even number then number of handshakes then will be the even or odd? If K is odd then will the number of handshakes that takes place be even or odd? 5

2. a) A circle of radius 1 is inscribed in an equilateral triangle of suitable size. Then three more circles are inscribed between the first circle and two sides of the triangle near each vertex. The process continues indefinitely, with progressively smaller circles. What is the sum of the radii of all the circles? 7

   b) Explain why it is impossible to have a polyhedron with 6 triangular faces meeting at each vertex. 8

3. a) Separate a standard deck of 52 card into five piles. What is the probability that one of the top card is face card (King, Queen or jack). 8

   b) Examine the equations: Determine the pattern and prove the identity. 7

   1=1

   2+3+4=1+8

   5+6+7+8+9=8+27

   10+11+12+13+14+15+16=27+64

4. a) Draw an 8*8 magic square such that each row, each column and each diagonal add up to the same number. Explain in detail the process you used to draw such 8*8 magic square. 7

OR

Solve the equation:

$$(4^x + 4^{-x}) - 54(2^x + 2^{-x}) + 101 = 0$$

b) Suppose that you have 12 marbles, all appearing the same but with one having an odd weight. You do not know whether the odd marble is heavier or lighter. How many weighing's are needed to find the odd marble? Describe briefly. 8

5. a) A Martini is made by mixing k parts gin with 1 part vermouth. Gin is usually 40% alcohol while vermouth 20% alcohol. A martini is said to be dry if it contain relatively vermouth. For instance, if k =15 then the martini is said to be dry. If instead k = 5 then the martini is said to be sweet. 8

b) There is two married couple that needs to cross the river. A small boat is available that will hold just two people at a time. The males involved are quite jealous. No woman can be left with a man unless her husband is also present. There are no other constraints. How can these four people cross the river? What is the fewest number of trips possible? 7

6. a) You have a piece of paper with a circle of radius between 2cm and 4cm drawn on it. You also have a plastic square of side 10cm. You have no ruler and no compass. How can you find the centre of the circle? Describe briefly. 5

b) Suppose a triangle with a base QR. Given a constructive method for drawing a segment parallel to QR, terminating on the sides PQ and PR of the triangle, that divides the triangle into two equal areas. 5

c) Suppose that S is a set with K element. Show that s has precisely 2^k subsets by induction. 5

7. Write short notes on: (**Any two**) 2×5

a) Why $11^{10} - 1$ is divisible by 100?

b) Which is greater, $\sin(\cos x)$ or $\cos(\sin x)$?

c) Show that the angle subtended by semi-circle is right angle triangle.